



Evolving Perspectives on Generative AI in Higher Education: Insights from Two Years of University Stakeholder Analysis

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Abstract

Technology innovations play an important role in urging higher education institutions to adapt to changing societal values and needs, yet the adoption of new technologies can be a complex process requiring multi-stakeholder engagement. This study examines the evolving perceptions of Generative AI (GenAI) within a university community through the analysis of survey data collected over two consecutive years. Through Latent Class Analysis, we identify different groups of perceptions among students, faculty, and staff. Findings highlight the role of experience with AI in shaping participants' attitudes. While faculty may recognize more of the values in GenAI applications, the level of comfort with using AI in education settings still varies. Meanwhile, students express persistent concerns related to job security, academic integrity, and ethical AI use. The study highlights the need for institution-wide AI literacy initiatives, adaptive learning frameworks, and structured policy interventions to support equitable and effective GenAI integration.

Keywords Higher education · Generative artificial intelligence · Technology acceptance · Innovation diffusion

Introduction

The integration of new technologies into education can be a complex and spiral process accompanied by both optimism and skepticism. Historically, the introduction of calculators in mathematics education provides one illustrative example: initial resistance centered on fears that calculators would erode students' basic arithmetic skills (Banks, 2011). Yet over time, educators came to view them as tools for enhancing problem-solving and conceptual understanding (Brown et al., 2007). This shift was shaped by professional development, curricular reforms, public perceptions, and broader cultural changes (Ruthven, 1995, 1996; Schmidt, 1998). More recently, breakthroughs in Generative AI (GenAI), in particular the public release of ChatGPT, have sparked a

fervent and almost unprecedented round of discussion about AI in society. Although applications of AI in education (AIED) have been receiving attention for their potential to assist administration, instruction, and learning (Chen et al., 2020), GenAI still came as a surprise to many educators. As Costa and Murphy, (2025) caution, drawing a direct parallel between calculators and GenAI risks oversimplification. Unlike calculators, large language models (LLMs) such as ChatGPT are capable of processing complex textual information and synthesizing cross-domain knowledge to respond to user inquiries (Brown et al., 2020; Stokel-Walker & Van Noorden, 2023). It has been reported that GPT can achieve human-level performance on various professional and academic exams, with scores comparable to those of top human test takers (OpenAI et al., 2024). Therefore, GenAI tools do not merely automate routine tasks; they generate language, synthesize information across domains, and influence how knowledge itself is produced and validated.

Despite unsolved issues with AI hallucination (Athaluri et al., 2023; Sok & Heng, 2023) and ongoing debates on AI plagiarism (Cotton et al., 2023), it seems inevitable that students and university employees will use GenAI tools at least personally (Baytas & Ruediger, 2024). In January 2025, the release of DeepSeek, a new AI model developed by a Chinese startup, has rocked people once again and upended

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several fundamental assumptions about AI progress (Roose, 2025). The surprising rapid development of AI urges Higher Education Institutions (HEIs) to make timely adaptations in their technology integration strategies, which should not only address immediate needs but also enhance the capabilities to respond to future technological innovations. Therefore, HEIs and their leadership are increasingly interested in evaluating the impact of GenAI and developing effective policies that govern its usage (McDonald et al., 2024; Wu et al., 2024). Recent scholarships suggest that many university AI policies tend to be fragmented or insufficient. For example, in a critical review of the world's top 20 universities, Luo, (2024) highlights that institutional GenAI policies are often reactive and lack coherence. A survey of top U.S. universities also found that fewer than half of the institutions provide detailed assessment policies, and only a minority offer guidance for faculty and researchers (An et al., 2025). It is emphasized that the next generation of GenAI policies must be human-centered, inclusive, and responsive to critical stakeholder perspectives (Holmes & Miao, 2023; Kim et al., 2025).

The literature has illustrated a wide range of perceived benefits and concerns of GenAI (Chan & Hu, 2023; Liu et al., 2024; Petricini et al., 2023). However, investigations mainly rely on cross-sectional designs, offering a snapshot of opinions at a single point in time. While one-time surveys or interviews provide useful baseline information, people's attitudes may change as they gain more exposure to AI technologies and respond to new institutional policies. The rapid development of AI itself further complicates this landscape, as improvements in AI capabilities can alter users' perceptions. In addition, previous studies focused on profiling students' or faculty's attitudes towards GenAI as a group, with limited investigations of potential subgroups and their differences. The current study seeks to address this gap by analyzing survey data collected over two consecutive years from the same institution, providing a unique opportunity to illustrate how the community's attitudes toward GenAI may have shifted or remained stationary during the time period. Through this comparison, this research aims to answer critical questions:

- 1) Whether and how do perceptions of GenAI in the university community form groups?
- 2) Whether and how do these perceptions evolve over time?

Related Work

Perceptions of GenAI in Higher Education

Implementing GenAI successfully requires a comprehensive understanding of the perceptions, concerns, and expectations of key stakeholders of higher education, especially students

and faculty. Across studies, students commonly report benefits such as help with idea generation, writing support, and clarifying difficult concepts, alongside persistent worries about accuracy, bias, privacy, and academic integrity (Chan & Hu, 2023; Liu et al., 2024). A nationally representative survey conducted in the UK found that 92% of undergraduates now use GenAI, mainly to explain concepts and summarize readings, while concerns about hallucinations and being accused of misconduct remain widespread and patterned by gender and socio-economic status (Freeman, 2025). At the same time, emerging work on trust indicates that students' reliance on GenAI depends on calibrated trust rather than simple enthusiasm. An exploratory survey of 253 students found heterogeneous trust levels that correlate with motivation and confidence, emphasizing the need to teach "trust calibration" rather than mere tool use (Amoozadeh et al., 2024).

Faculty perspectives are similarly nuanced. Multi-institution interview studies show many instructors are experimenting with GenAI for AI-literacy activities yet continue to voice concerns about assessment validity, research quality, and uneven institutional support (Baytas & Ruediger, 2025). In one U.S. survey with 982 students and 76 faculty, the two groups reported broadly similar attitudes overall, diverging mainly on ease/fun of use and perceived effects on grades, with notable differences across gender and STEM vs. non-STEM majors (Kim et al., 2025). Finally, several reviews and discipline-focused studies converge on academic-integrity risks and equity issues as cross-cutting concerns that institutions must address through clearer policies and targeted support (Ahmed et al., 2024; Kofinas et al., 2025; Mamo et al., 2024; Petricini et al., 2023). Together, these studies illustrate a wide range of perceived benefits and concerns, highlighting the need to further investigate underlying factors that shape people's perceptions.

Diffusion of Innovations in Higher Education

The Diffusion of Innovations (DOI) framework is one of the most influential lenses for understanding how new practices spread in organizations and social systems (Greenhalgh et al., 2004; Rogers, 2003). DOI explains adoption as a function of perceived relative advantage, compatibility, complexity, trialability, and observability, unfolding through communication channels, decision structures, and social-system norms (Rogers, 2003; Sahin, 2006; Wisdom et al., 2014). A core contribution of DOI is its account of adopter heterogeneity. Innovations typically diffuse through five adopter categories distinguished by motivations, risk tolerance, and communication behavior: 1) Innovators: risk-takers who are the first to adopt and experiment with new technologies; 2) Early Adopters: influential individuals who quickly recognize the value of innovations and advocate for their

adoption; 3) Early Majority: a pragmatic group that adopts innovations after observing their success and viability; 4) Late Majority: more skeptical individuals who adopt innovations once they are widely accepted and proven effective; 5) Laggards: conservative individuals who resist adoption until the innovation becomes unavoidable.

Applications of DOI theory in higher education confirm that technology acceptance is stratified across adopter categories and that effectiveness depends on tailoring implementation to those segments. Systematic review evidence shows academics' uptake of learning technologies varies with perceived innovation attributes, user characteristics, and institutional context (Liu et al., 2020). Comparative studies of the "reluctant majority" versus early adopters find that mainstream faculty require stronger pedagogical value propositions, clearer incentives, and structured professional development, cautioning against extrapolating from early-adopter enthusiasm (Hixon et al., 2012). At the institutional level, multi-case analyses of blended-learning initiatives show that support must evolve as adoption moves from innovators/early adopters to the early and late majority (Porter et al., 2014, 2016). Collectively, the literature highlights the importance of understanding adopter categories and designing differentiated support mechanisms to successfully advance innovation diffusion in higher education.

Methods

Instruments and Data Collection

The survey items were collaboratively developed by the research team through an iterative process that included reviewing emerging literature on GenAI in education, consulting with faculty and educational technology experts, piloting drafts for tests, and revising for clarity and relevance. This approach ensured that the instrument

reflected diverse stakeholder perspectives and addressed issues salient to the institutional context. While the items were designed for exploratory purposes rather than drawn directly from existing theoretical frameworks, the research team tested their readability and empirical validity through pilot testing and expert reviews. The first-year questionnaire contained six groups of closed-ended items. Summarizing descriptions of survey items in each group are listed in Table 1. All items are framed in a Likert scale format, where respondents are requested to rate their level of agreement with the list of statements from **Strongly Disagree** to **Strongly Agree**.

For the Year 2 Questionnaire, survey items were adjusted to reflect the fast-evolving understanding of AI technologies and changes in the institution and study context. For example, instead of using the statement "I have received instruction about how to use ChatGPT or similar tools", it was changed to "I have received **sufficient training** about how to use generative AI". More importantly, the study was initially designed to examine only student and faculty perceptions in Year 1. After completing Year 1 data collection, however, staff members expressed a strong interest in participating. In response, we expanded the survey scope in Year 2 to include staff as potential participants and standardized the wording of survey items to improve comparability of results across students, faculty, and staff. The summarizing descriptions of the Year 2 questionnaire are listed in Table 2. Participants only receive items applicable to their primary affiliation with the university.

The study was conducted at a public, research-intensive university in the mid-Atlantic region of the United States and approved by the institution's IRB office. Recruitment materials containing a link to the questionnaire were sent to faculty, students, and staff via their university email addresses. Participants volunteered to participate in the study and respond to the questionnaires online. The Year 1

Table 1 Survey items groups for year 1 questionnaire

Group	Description	Number of Items
Experience and Training with AI (E)	Self-reported levels of familiarity, experience, and training with GenAI	Students: 5 Faculty: 5
Prevalence of AI (PR)	Perception of how prevalent AI usage is in education	Students: 3 Faculty: 2
AI Policies (P)	Attitudes towards regulating or restricting AI usage	Students: 3 Faculty: 4
Perception of Successful AI Usage (S)	Perception of how successful AI is used by themselves or others	Students: 4 Faculty: 2
Comfort with GenAI usage (C)	Levels of comfort with AI usage in academic settings	Students: 3 Faculty: 6
Value of AI (V)	Perception of the potential positive and negative value of AI	Students: 4 Faculty: 3

Table 2 Survey items groups for year 2 questionnaire

Group	Description	Number of Items
Experience and Training with AI (E)	Self-reported levels of familiarity, experience, and training with GenAI	3
Risks and Concerns of AI (R)	Perception of potential risks associated with AI	4
AI Policies (P)	Attitudes towards classroom and university-level AI policies	4
Academic Integrity (A)	Perception of academic integrity issues associated with AI usage	4
Perception of Successful AI Usage (S)	Perception of how successful AI is used by themselves or others	6
Comfort with GenAI usage (C)	Levels of comfort with AI usage in academic settings	5

survey was open from March to May in 2023, and the Year 2 survey was open from March to May in 2024, allowing sufficient time for participation.

It should be noted that this study was conducted at a single institution, and its specific policy and environment may not be representative of the broader higher education landscapes. As context, between 2023 and 2025, the university's stance on GenAI seems to shift from a primarily prohibitive and integrity-focused approach to a more nuanced, flexible framework. For example, in 2023, the sample syllabus statement it provided emphasized academic misconduct, warning that “you must complete all course work entirely on your own” unless explicitly permitted, framing AI use largely as a potential violation with “severe consequences”. In 2025, the updated sample statement has recognized diverse instructional goals, offering instructors multiple options ranging from full prohibition to conditional or process-focused use, noting that “students may use generative AI tools to brainstorm ideas, to create an outline... and to improve their use of grammar,” provided they cite their tools.

Data Analysis

Data analyses were performed using R and R Studio. Participants' responses to all Likert Scale items were converted to numeric values based on the five-point scale: 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree. Latent Class Analysis (LCA, Vermunt & Magidson, 2002) was used to identify unobserved subgroups (latent classes) within the survey respondents based on their response patterns. This method allows for the classification of individuals into mutually exclusive groups based on their shared characteristics, as inferred from the observed data (Collins & Lanza, 2009). A series of models with increasing numbers of latent classes were estimated using the *poLCA* package (Linzer & Lewis, 2011), and model fit was assessed using the Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC). The determination of the optimal number of classes was guided by model fit indices, with lower BIC and AIC values indicating better model fit. The final number of latent classes was chosen based on the fit

indices and practical interpretability. Models for students, faculty, and staff in each year were estimated and evaluated separately.

After identifying the latent classes through LCA, further statistical tests were conducted to examine differences in the mean responses across the identified classes. To account for unequal variances between classes and the ordinal nature of the responses, Welch's ANOVA was used to test for differences in mean responses across the latent classes. Following a significant Welch ANOVA result, Dunn's post hoc test with Bonferroni correction was applied to perform pairwise comparisons between the latent classes. The significance level for all tests was set at $\alpha = 0.05$. The results of these tests were used to identify specific pairwise differences in the mean responses across latent classes.

Results

In Year 1, we collected 590 valid survey responses in total, which contained 334 responses from students and 256 from faculty. In Year 2, we collected 364 valid survey responses in total, which contained 80 responses from faculty, 219 from staff, and 65 from students. The average ratings of each survey item are presented in Tables 3, 4 and 5. The results of LCA are presented in the following sections.

Latent Classes in Year 1

Student Classes

For the 334 students in Year 1, LCA suggests that three classes exist in the data. The characteristics of each class are illustrated in Fig. 1. The height of each bar represents the average response value to a survey item in the class. The white break line indicates that the average has reached the neutral value of 3, so a value above three would indicate “agreement”, and vice versa, a value below three would indicate “disagreement”. The percentage of each class among the respondents is also indicated in the center. The post-hoc analysis showed that the three classes have significant

Table 3 Year 1 student survey items and average ratings

ID	Item	Mean
E1	I am familiar with ChatGPT	3.47
E2	I am familiar with the concept of artificial intelligence (AI)	4.04
E3	I have experience using ChatGPT	2.62
E4	I have received instructions about how to use ChatGPT or similar tools	1.62
E5	I would be open to receiving instruction about how to use ChatGPT or similar tools	3.49
PR1	Students' use of AI text generation tools to complete coursework is inevitable	3.81
PR2	Student use of AI text generation tools to complete coursework is prevalent in higher education	3.15
PR3	The use of AI in education is very prevalent	3.12
P1	Students should not be restricted from using AI for coursework	2.90
P2	Something must be done to stop the use of AI by students	2.94
P3	Using generative AI tools to complete coursework violates the university's academic integrity policies	3.84
S1	My instructors have addressed the use of AI (especially ChatGPT and other text and image generation tools) in my courses	2.49
S2	My instructors have integrated AI generators like ChatGPT into their instruction	1.51
S3	Instructors misuse AI in academic settings	2.75
S4	Instructors use AI well in academic settings	2.80
C1	I trust AI to grade my assignments and assessments for my courses instead of my instructor	2.11
C2	I would feel confident knowing an instructor was using an AI-created syllabus	2.53
C3	I plan to use ChatGPT or similar tools for my coursework in the future	2.28
V1	Artificial Intelligence has value in education	3.78
V2	AI is used in education for good and helpful reasons	3.38
V3	AI is misused in education	3.39
V4	Artificial intelligence (in the form of text and image generation) could be dangerous for students	3.40

differences across several survey items, and the statistics are presented in Table 6.

The three classes are named Advocates, Explorers, and Skeptics based on their profiles, which are described below:

Class 1 (Advocates): The Advocates are most experienced with AI and show the most positive attitude about its usage in education. This class makes up 34% of the student respondents and features a high level of familiarity with the concept of ChatGPT (E1) and AI (E2), as well as a moderate level of experience with ChatGPT (E3), with average ratings all exceeding 3. They are also highly open to receiving more instructions about AI (E5). This class shows a preference for less restrictive AI policies by agreeing that students should not be restricted from using AI (P1) and disagreeing that the university needs to stop student use of AI (P2). Additionally, these students perceive the value of AI in education (V1) and that AI is used for good and helpful reasons (V2).

Class 2 (Explorers): The Explorers (16% of students) exhibit a more neutral stance. The profile is similar to that of Advocates in some aspects, featuring a high level of familiarity with AI, disagreement toward restrictive AI policies, and neutral attitudes toward the use of AI in education settings. While Explorers still perceive good

values in AI, the ratings are also significantly lower than those of Advocates, suggesting a more balanced view in comparison. Interestingly, these students have received more instructions about AI (E4), and their instructors have addressed and integrated AI into courses to a certain extent (S1 and S2), which may have contributed to their balanced perceptions.

Class 3 (Skeptics): Skeptics (50% of students) have significantly lower levels of familiarity with AI (E1–E3). These students report a mediocre interest in receiving instructions (E5) and agree that student use of AI should be restricted (P1 and P2) and that using AI in coursework would violate academic integrity policy (P3). Not only do they feel uncomfortable using AI for coursework (C3), but they also prefer instructors not to use AI for grading assignments (C1) or creating syllabi (C2). Compared with their neutral opinions on the positive value of AI (V1 and V2), they perceive more danger and misuse of AI in education (V3 and V4).

Faculty Classes

For faculty respondents, we also identified three latent classes, and their profiles are illustrated in Fig. 2. These profiles are consistent with those of students, aligning closely

Table 4 Year 1 faculty survey items and average ratings

ID	Item	Mean
E1	I am familiar with the concept of artificial intelligence (AI)	3.70
E2	I am familiar with ChatGPT	4.18
E3	I have experience using ChatGPT	2.51
E4	I have participated in training (or related faculty development) on the use of ChatGPT (and/or similar AI text and image generators)	1.57
E5	I am interested in receiving training (or related faculty development) on the use of ChatGPT (and/or similar AI text and image generators)	3.49
PR1	Student use of AI text-generation tools to complete coursework is inevitable	3.99
PR2	Student use of AI text-generation tools to complete coursework is prevalent in higher education	3.25
P1	Students should be restricted from using AI text-generation tools in their coursework	3.30
P2	Students will need to be taught how to use AI text-generation tools appropriately	4.32
P3	Student use of AI text-generation tools to complete coursework is cheating	3.66
P4	Using AI text-generation tools to complete coursework violates academic integrity policies at the University	3.66
S1	I have addressed the use of AI generators such as ChatGPT with my students	2.34
S2	I integrate ChatGPT (or similar AI text and image generators) in my instruction	1.56
C1	I plan to integrate ChatGPT (or similar tools) into my instruction in the future	2.38
C2	I would be comfortable using an AI text generator to build a course syllabus	2.45
C3	I would be comfortable using an AI text generator to build a course assignment or activity	2.68
C4	I would be comfortable using an AI program to grade assignments in my courses	2.30
C5	I intend to use an AI text-generation tool as part of my research, scholarship, or creative activities	2.43
C6	I intend to use a different AI tool (not a text or image generator) as part of my research, scholarship, or creative activities	2.50
V1	The use of AI text-generation tools could be beneficial for students	3.39
V2	Artificial intelligence (in the form of text and image generation) could be dangerous for students	3.64
V3	Students misuse AI text generation tools	3.59

with Advocates, Explorers, and Skeptics. Post-hoc analysis of group comparisons is presented in Table 7.

Class 1 (Advocates): Advocates, who make up 27% of the faculty, are familiar and experienced with AI (E1–E3) but also highly interested in receiving further training (E5). They reported a significantly lower agreement to statements for restricting students' use of AI (P1) or considering using AI in coursework as cheating (P3) and a violation of academic integrity (P4). Nonetheless, they agree that students will need to be taught how to use AI tools appropriately (P2). These faculty are also comfortable with using AI for various teaching and research activities (C1 – C6) and are likely to perceive the benefits AI may bring to students (V1). They lean towards neutrality regarding the potential dangers and misuse of AI (V2 and V3).

Class 2 (Explorers): Compared to Advocates, Explorers (52% of the faculty) are less experienced with ChatGPT (E3) and moderately agree that using AI in coursework violates academic integrity policies. They still agree that it is necessary to teach students how to use AI (P2). Their level of comfort with using AI in various educational contexts is below the neutral value. And their perceptions of

the benefits and dangers of using AI are balanced, both slightly above the neutral value.

Class 3 (Skeptics): Skeptics (21% of the faculty) are familiar with AI concepts (E1 and E2) but relatively inexperienced with ChatGPT (E3) and lack interest in receiving more instructions (E5). These faculty also display strong agreement that students' use of AI should be restricted (P1) and that using AI in coursework is cheating (P3). They are unlikely to have addressed AI issues or integrated AI tools into instructions (S1 and S2). Their level of comfort with using AI is significantly lower than the other two classes (C1–C6), and they perceive more dangers and misuse associated with AI (V2 and V3) than benefits (V1).

Latent Classes in Year 2

Student Class

For the 65 students in Year 2, LCA suggests that only one class exists among the respondents, and the average ratings of survey items are illustrated in Fig. 3. These students are experienced with using AI (E1) but do not think they have received sufficient training (E2) and are willing to receive

Table 5 Year 2 survey items and average ratings

ID	Item	Faculty	Staff	Student
E1	I'm experienced with using generative AI	3.36	3.10	4.03
E2	I have received sufficient training about how to use generative AI	2.67	2.13	2.29
E3	I'm willing to receive more instructions about how to use generative AI	4.26	4.17	3.49
R1	Academic integrity issues are a risk of using generative AI	4.33	4.13	4.06
R2	Advancing bias and stereotypes is a risk of using generative AI	3.78	3.79	3.50
R3	Privacy violations is a risk of using generative AI	3.49	3.86	3.47
R4	I fear my job will be replaced by generative AI in the future	2.33	2.47	4.06
P1	The University needs clearer policies or rules about using AI as a student	4.10	4.28	3.72
P2	I/My instructors have a clear policy about students' use of generative AI in my class	3.59		4.03
P3	The University needs clearer policies or rules about using AI as a faculty/staff	3.89	4.21	
P4	I will use generative AI regardless of university policy	2.66	2.61	2.57
A1	Using generative AI tools to complete coursework violates academic integrity policies at the University	3.35	3.46	3.72
A2	AI detection tools are accurate (i.e., Turnitin, GPTZero)	2.74		
A3	I worry that my instructor may blame me for using generative AI on assignments or exams even if I didn't			4.25
A4	My instructor will trust me when I say I didn't use generative AI			3.34
C1	I will (or will continue to) use generative AI for my coursework/instruction in the future	3.76		2.79
C2	I'm comfortable using generative AI as part of my research, scholarship, or creative activities	3.02		
C3	I'm comfortable using generative AI to build a course assignment or activity	2.94		
C4	[I feel comfortable]/[I feel comfortable knowing an instructor is] using an AI-generated syllabus	2.71		2.53
C5	[I feel comfortable using AI in]/[I trust generative AI in] grading my assignments and assessments	2.20		2.20
S1	I am confident in my instructors' use of generative AI in the classroom			2.77
S2	My instructors have successfully used generative AI in their instruction			2.49
S3	My instructors practice generative AI use responsibly in the classroom			3.00
S4	Students misuse generative AI	3.78	3.68	
S5	Students should be informed when instructors use generative AI for instructional support	3.43	3.88	4.42
S6	We need to teach students more about using generative AI responsibly	4.61	4.58	

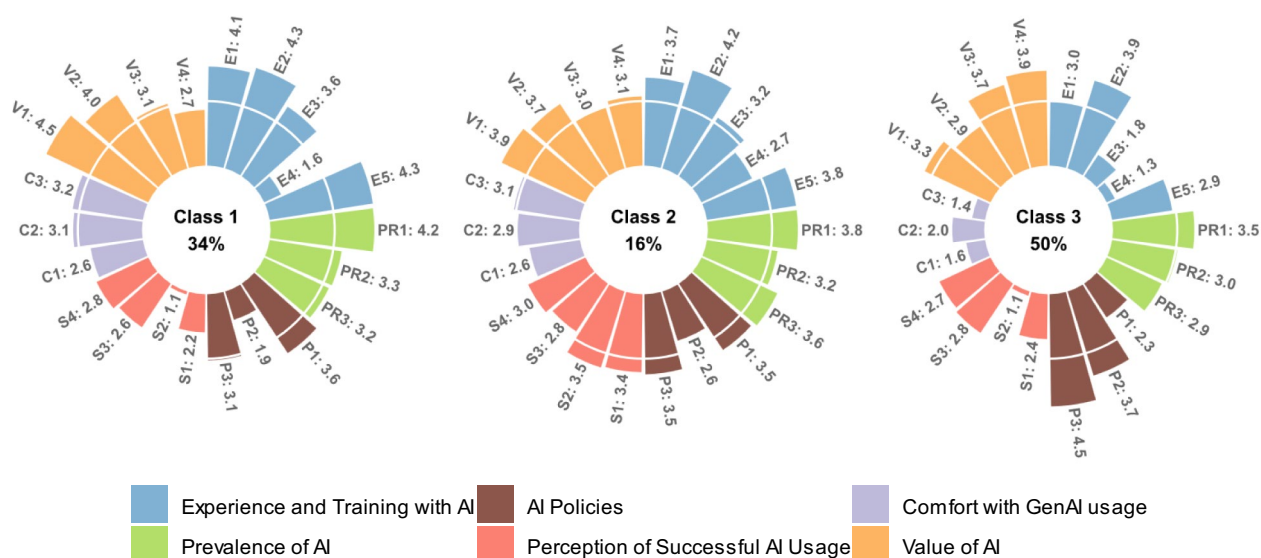
**Fig. 1** Student latent class characteristics in year 1

Table 6 Dunn's test of multiple comparisons of year 1 student latent classes (statistical significance: ns=not significant, *=p<0.05, **=p<0.01, ***=p<0.001, ****=p<0.0001)

ID	Comparison	z-value	p-value	Significance
E1	Class 1–2	−1.73	.084	ns
	Class 1–3	−6.15	<.001	****
	Class 2–3	−3.02	.005	**
E2	Class 1–2	−0.91	.363	ns
	Class 1–3	−3.21	.004	**
	Class 2–3	−1.57	.235	ns
E3	Class 1–2	−1.38	.168	ns
	Class 1–3	−8.42	<.001	****
	Class 2–3	−5.19	<.001	****
E4	Class 1–2	6.38	<.001	****
	Class 1–3	−2.53	.011	*
	Class 2–3	−8.76	<.001	****
E5	Class 1–2	−2.70	.007	**
	Class 1–3	−8.98	<.001	****
	Class 2–3	−4.21	<.001	****
PR1	Class 1–2	−2.82	.010	**
	Class 1–3	−5.03	<.001	****
	Class 2–3	−0.96	.336	ns
PR2	Class 1–2	−0.44	.658	ns
	Class 1–3	−1.86	.188	ns
	Class 2–3	−0.99	.640	ns
PR3	Class 1–2	1.99	.047	*
	Class 1–3	−2.74	.012	*
	Class 2–3	−4.25	<.001	****
P1	Class 1–2	−0.49	.625	ns
	Class 1–3	−8.61	<.001	****
	Class 2–3	−6.25	<.001	****
P2	Class 1–2	3.15	.002	**
	Class 1–3	11.38	<.001	****
	Class 2–3	5.61	<.001	****
P3	Class 1–2	1.73	.084	ns
	Class 1–3	9.36	<.001	****
	Class 2–3	5.50	<.001	****
S1	Class 1–2	4.99	<.001	****
	Class 1–3	0.91	.364	ns
	Class 2–3	−4.57	<.001	****
S2	Class 1–2	13.69	<.001	****
	Class 1–3	0.17	.864	ns
	Class 2–3	−14.38	<.001	****
S3	Class 1–2	1.34	.364	ns
	Class 1–3	2.04	.124	ns
	Class 2–3	0.19	.850	ns
S4	Class 1–2	1.89	.116	ns
	Class 1–3	−0.40	.687	ns
	Class 2–3	−2.32	.061	ns
C1	Class 1–2	0.04	.968	ns
	Class 1–3	−7.09	<.001	****
	Class 2–3	−5.61	<.001	****

Table 6 (continued)

ID	Comparison	z-value	p-value	Significance
C2	Class 1–2	−0.93	.351	ns
	Class 1–3	−7.62	<.001	****
	Class 2–3	−5.00	<.001	****
C3	Class 1–2	−0.25	.801	ns
	Class 1–3	−10.99	<.001	****
	Class 2–3	−8.40	<.001	****
V1	Class 1–2	−3.74	<.001	***
	Class 1–3	−9.72	<.001	****
	Class 2–3	−3.68	<.001	***
V2	Class 1–2	−2.01	.044	*
	Class 1–3	−9.16	<.001	****
	Class 2–3	−5.07	<.001	****
V3	Class 1–2	−0.63	.529	ns
	Class 1–3	4.88	<.001	****
	Class 2–3	4.50	<.001	****
V4	Class 1–2	1.96	.050	*
	Class 1–3	8.15	<.001	****
	Class 2–3	4.32	<.001	****

more instructions (E3). They believe that using AI to complete coursework violates academic integrity policies (A1) and are concerned that instructors may wrongfully blame them for using AI (A3). Still, they slightly agree that “My instructor will trust me when I say I didn't use generative AI (A4).” While these students report that their instructors have a clear policy regarding AI usage (P2), they also think that there is a need for clearer university-level policies (P1). Regarding the perception of successful AI usage, they slightly disagree or are unsure whether instructors use AI successfully or responsibly (S1–S3), but agree that students misuse GenAI (S5). Similar to the results of Year 1, students are not yet comfortable with using AI in classroom settings (C1–C5). Students show agreement with all four risks associated with AI, especially those related to academic integrity (R1) and job security (R4). Overall, the student profile in Year 2 may be called **Experienced Explorers**, with a high level of experience with AI and evident concerns.

Faculty Classes

Two classes are identified for the 80 faculty respondents. The characteristics of each class are illustrated in Fig. 4 and the post-hoc analysis of class differences is presented in Table 8. We name these **Advocates** and **Skeptics** based on their characteristics as described below.

Class 1 (Advocates): Similar to Advocates in Year 1, these faculty members (43%) feature a high level of experience with GenAI usage (E1), have received a

Table 7 Dunn's test of multiple comparisons of year 1 faculty latent classes (statistical significance: ns=not significant, *= $p<0.05$, **= $p<0.01$, ***= $p<0.001$, ****= $p<0.0001$)

ID	Comparison	z-value	p-value	Significance
E1	Class 1–2	−4.65	<.001	***
	Class 1–3	−3.00	.005	**
	Class 2–3	0.86	.389	ns
E2	Class 1–2	−6.84	<.001	***
	Class 1–3	−4.19	<.001	***
	Class 2–3	1.51	.130	ns
E3	Class 1–2	−8.01	<.001	***
	Class 1–3	−6.73	<.001	***
	Class 2–3	−0.26	.795	ns
E4	Class 1–2	−5.79	<.001	***
	Class 1–3	−6.00	<.001	***
	Class 2–3	−1.45	.146	ns
E5	Class 1–2	−3.21	.003	**
	Class 1–3	−5.08	<.001	***
	Class 2–3	−2.77	.006	**
PR1	Class 1–2	−4.78	<.001	***
	Class 1–3	−2.96	.006	**
	Class 2–3	1.03	.301	ns
PR2	Class 1–2	−1.96	.151	ns
	Class 1–3	0.12	.902	ns
	Class 2–3	1.92	.151	ns
P1	Class 1–2	6.22	<.001	***
	Class 1–3	11.37	<.001	***
	Class 2–3	7.09	<.001	***
P2	Class 1–2	−1.24	.216	ns
	Class 1–3	−4.44	<.001	***
	Class 2–3	−3.85	<.001	***
P3	Class 1–2	5.80	<.001	***
	Class 1–3	11.07	<.001	***
	Class 2–3	7.14	<.001	***
P4	Class 1–2	6.11	<.001	***
	Class 1–3	10.89	<.001	***
	Class 2–3	6.65	<.001	***
S1	Class 1–2	−5.78	<.001	***
	Class 1–3	−4.61	<.001	***
	Class 2–3	0.10	.919	ns
S2	Class 1–2	−7.72	<.001	***
	Class 1–3	−7.04	<.001	***
	Class 2–3	−0.86	.391	ns
C1	Class 1–2	−6.65	<.001	***
	Class 1–3	−10.08	<.001	***
	Class 2–3	−5.25	<.001	***
C2	Class 1–2	−5.65	<.001	***
	Class 1–3	−9.10	<.001	***
	Class 2–3	−5.06	<.001	***
C3	Class 1–2	−6.49	<.001	***
	Class 1–3	−10.19	<.001	***
	Class 2–3	−5.53	<.001	***

Table 7 (continued)

ID	Comparison	z-value	p-value	Significance
C4	Class 1–2	−2.47	.013	*
	Class 1–3	−7.32	<.001	***
	Class 2–3	−5.96	<.001	***
C5	Class 1–2	−7.76	<.001	***
	Class 1–3	−11.26	<.001	***
	Class 2–3	−5.57	<.001	***
C6	Class 1–2	−5.47	<.001	***
	Class 1–3	−9.76	<.001	***
	Class 2–3	−5.97	<.001	***
V1	Class 1–2	−6.28	<.001	***
	Class 1–3	−10.77	<.001	***
	Class 2–3	−6.37	<.001	***
V2	Class 1–2	1.96	.050	ns
	Class 1–3	7.58	<.001	***
	Class 2–3	6.72	<.001	***
V3	Class 1–2	1.49	.178	ns
	Class 1–3	2.72	.019	*
	Class 2–3	1.70	.178	ns

slight degree of training about GenAI (E2), and are willing to receive more instructions about it (E3). These faculty do not perceive using AI in coursework as a violation of academic integrity policies (A1) and do not fully trust AI detection tools (A2). Similar to students, they believe that they have a clear AI policy in class (P2) but also desire clearer policies at the university level (P1), especially policies for faculty (P3). However, there is a possibility they will use GenAI regardless of the university policy (P4). They are almost neutral regarding whether students misuse GenAI (S4) and whether students need to be informed when instructors use AI (S5) but do strongly agree that the university needs to teach students about using AI responsibly (S6). They are generally comfortable with using GenAI for research (C2), building syllabi (C4), and course materials (C3), but not in grading assignments (C5). While they perceive AI's risks in causing academic integrity issues (R1) and advancing bias and stereotypes (R2), they do not fear that AI would replace their jobs (R4).

Class 2 (Skeptics): Skeptics (57% in the faculty) in Year 2 have a level of experience slightly below the neutral value (E1) and inadequate training (E2), but are open to receiving more instructions (E3). Their views on AI policies are similar to those of Advocates, except that they are unlikely to use GenAI regardless of the university policy (P4). They also report low levels of comfort with using AI in various professional activities (C2–C5). They also perceive the risks associated with AI and are more concerned about academic integrity

Faculty Latent Class Characteristics in Year 1

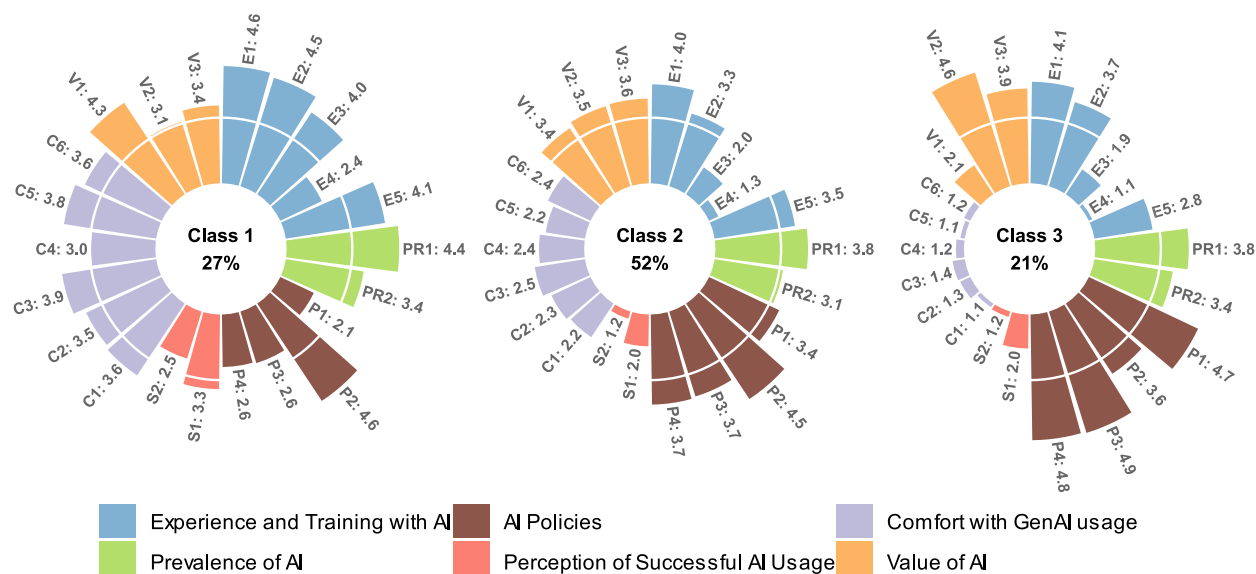
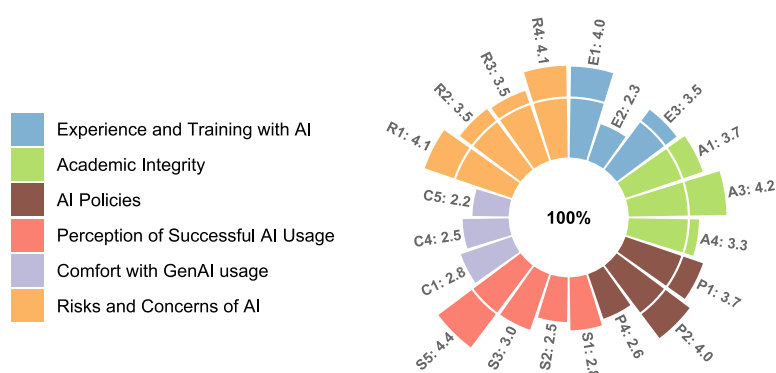


Fig. 2 Faculty latent class characteristics in year 1

Fig. 3 Student latent class characteristics in year 2

Student Latent Class Characteristics in Year 2



issues than Advocates (R1) and are neutral about the risk of AI replacing jobs (R4).

Staff Classes

Staff make up the largest population of the respondents in Year 2. Similar to faculty, staff could be categorized into two distinct classes, which were labeled as **Advocates** and **Skeptics**. Their profiles are illustrated in Fig. 5, and the post-hoc analysis of class differences is presented in Table 9.

Class 1 (Advocates): This class constitutes 66% of the staff and is experienced with using GenAI (E1) but has not received sufficient training about GenAI (E2) and

strongly agrees to receive more instructions about it (E3). They stay neutral about whether using AI in coursework violates academic integrity policies (A1). Meanwhile, they desire clearer AI policies at the university level (P1) but stay neutral about using AI regardless of the university policy (P4). They slightly agree that students misuse AI (S4) and that students should be informed about instructors' use of AI for instructional support (S5). They strongly agree that the university needs to teach students more about using AI responsibly (S6). These staff perceive all risks associated with AI except that AI may replace their jobs (R4). Overall, the high level of experience and strong interest in knowing more about AI qualify Class 1 as Advocates.

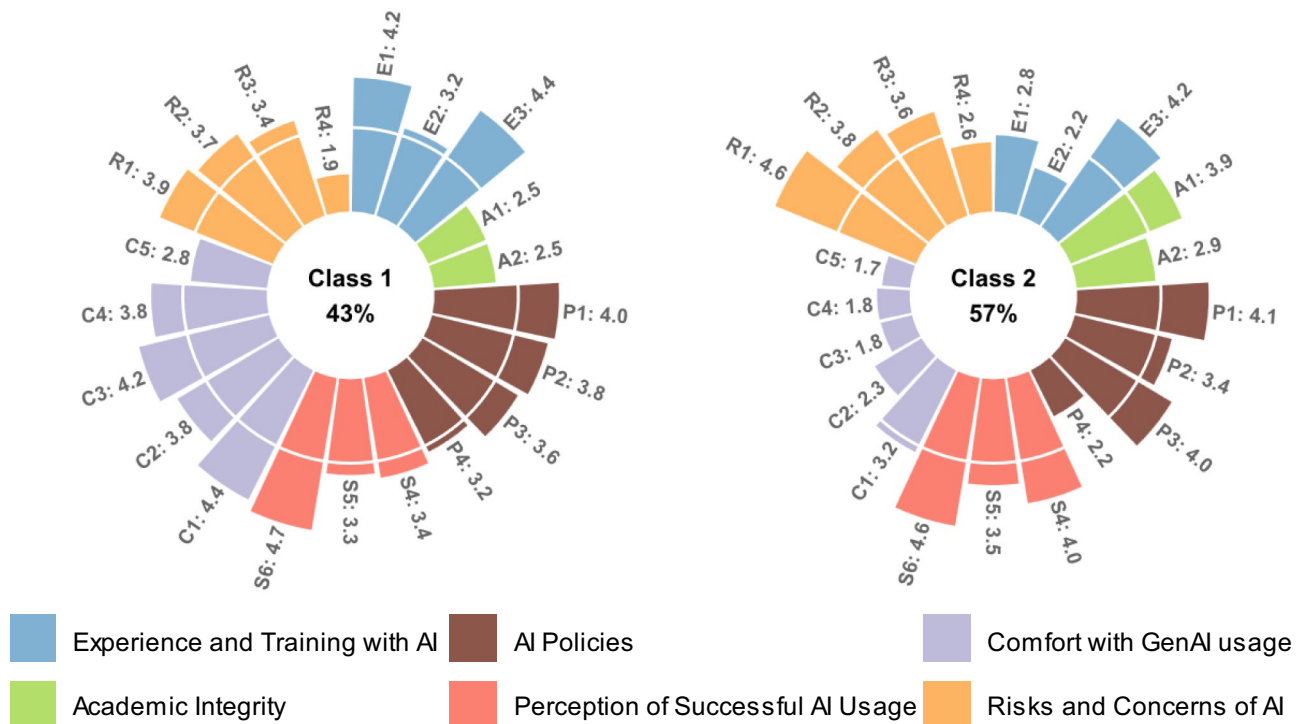


Fig. 4 Faculty latent class characteristics in year 2

Table 8 Dunn's test of multiple comparisons of year 2 faculty latent classes (statistical significance: ns=not significant, *= $p<0.05$, **= $p<0.01$, ***= $p<0.001$, ****= $p<0.0001$)

ID	Comparison	z-value	p-value	Significance
E1	Class 1–2	−5.22	<.001	****
E2	Class 1–2	−3.81	<.001	***
E3	Class 1–2	−0.70	.485	ns
R1	Class 1–2	3.31	<.001	***
R2	Class 1–2	0.57	.566	ns
R3	Class 1–2	1.06	.290	ns
R4	Class 1–2	2.76	.006	**
P1	Class 1–2	0.94	.349	ns
P2	Class 1–2	−1.41	.159	ns
P3	Class 1–2	0.97	.332	ns
P4	Class 1–2	−3.04	.002	**
A1	Class 1–2	5.27	<.001	****
A2	Class 1–2	1.33	.184	ns
C1	Class 1–2	−4.61	<.001	****
C2	Class 1–2	−4.54	<.001	****
C3	Class 1–2	−6.91	<.001	****
C4	Class 1–2	−5.94	<.001	****
C5	Class 1–2	−3.92	<.001	****
S4	Class 1–2	2.30	.022	*
S5	Class 1–2	0.60	.547	ns
S6	Class 1–2	−0.31	.758	ns

Class 2 (Skeptics): These staff (34%) have a significantly lower level of experience and training related to GenAI (E1 and E2). They are still willing to receive more instructions, but to a significantly lesser extent than Advocates (E3). They strongly believe that using AI in coursework violates academic integrity policies (A1). Their need for clearer AI policies is consistent with Advocates, but they are unlikely to use AI regardless of the university policy (P4).

Discussion

The Staged Adoption of New Technologies

Modern society is increasingly shaped by technology, as it pervades all aspects of modern life and influences how we interact with the world and each other (Ellul, 2021). Education, in particular, has witnessed a series of opportunities and challenges brought by technology innovations in the past decades, such as massive open online courses (Baturay, 2015), hybrid learning (Raes et al., 2020), and virtual reality (VR) applications (Freina & Ott, 2015). The successful adoption of new technologies has the potential to improve the accessibility of the learning environment (Alper & Raharinirina, 2006), enhance student engagement (Liu

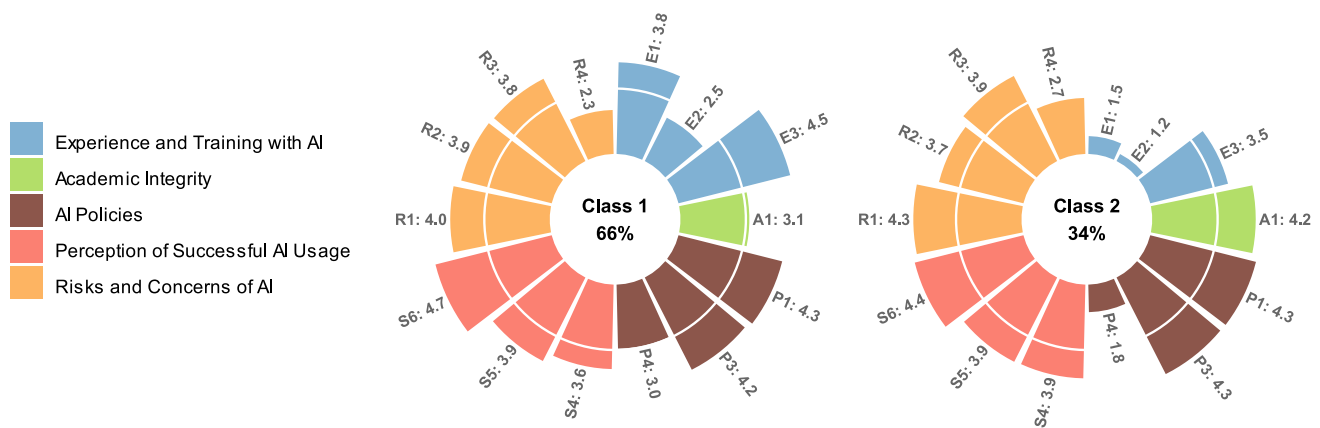


Fig. 5 Staff latent class characteristics

Table 9 Dunn's test of multiple comparisons of year 2 staff latent classes (statistical significance: ns=not significant, *= $p<0.05$, **= $p<0.01$, ***= $p<0.001$, ****= $p<0.0001$)

ID	Comparison	z-value	p-value	Significance
E1	Class 1–2	−10.43	<.001	****
E2	Class 1–2	−7.28	<.001	****
E3	Class 1–2	−6.63	<.001	****
R1	Class 1–2	2.41	.016	*
R2	Class 1–2	−1.10	.269	ns
R3	Class 1–2	0.91	.364	ns
R4	Class 1–2	1.89	.059	ns
P1	Class 1–2	0.61	.540	ns
P3	Class 1–2	1.22	.221	ns
P4	Class 1–2	−5.91	<.001	****
A1	Class 1–2	6.82	<.001	****
S4	Class 1–2	2.09	.036	*
S5	Class 1–2	0.29	.774	ns
S6	Class 1–2	−2.37	.018	*

et al., 2020), and eventually, improve education outcomes (Tamim et al., 2011). However, the implementation of new technologies in education is not always a smooth and obstacle-free process. In fact, a number of factors, such as faculty and student resistance, privacy and security concerns, and infrastructure and maintenance costs, could impact the adoption of technology in the educational context.

Previous studies have applied existing theoretical frameworks to model the process of technology adoption in higher education (Fathema et al., 2015; Pinho et al., 2021). These frameworks attempt to understand factors that impact the technology adoption process, not only from the individual perspective but also considering the organizational and societal dynamics. As an exploratory study, our survey items were empirically derived rather than theory-driven. Nonetheless, the survey content aligns closely with constructs in

widely used technology adoption frameworks, such as the Technology Acceptance Model (TAM) (Davis et al., 1989) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2016). For example, items capturing perceptions of the value and success of AI usage parallel the Perceived Usefulness construct in TAM and the Performance Expectancy dimension in UTAUT, both of which assess beliefs about technology's ability to enhance outcomes. Similarly, items on comfort with AI usage and experience/training reflect Perceived Ease of Use in TAM and Effort Expectancy in UTAUT, emphasizing the role of familiarity and confidence in reducing adoption barriers. Constructs such as AI policies, risks/concerns, and academic integrity align with UTAUT's Facilitating Conditions and Social Influence dimensions, highlighting organizational and societal factors that shape technology acceptance. Thus, even though the survey items were not explicitly theory-based, they could map onto established constructs, supporting their conceptual validity and situating our findings within the broader technology adoption literature.

More importantly, our results align with the DOI theory by identifying the existence of three latent classes in the sample population: Advocates who are experienced with GenAI and exhibit positive attitudes towards its application in education, Explorers who are aware of the risks of GenAI but are willing to learn about it, and Skeptics who are inexperienced with GenAI and concerned about its negative impact on education. We did not use the DOI categories (Innovators, Early Adopter, Early Majority, Late Majority, and Laggards) to label the classes mainly because of a widely critiqued limitation of the theory: the assumption that innovation is inherently beneficial. Rogers' broader treatment of innovations throughout the theory and especially the usage of the label "Laggards", tends to equate adoption with progress and resistance with backwardness, thus framing non-adoption as a deficiency rather than a potentially rational or context-sensitive decision (Wolfe, 1994). This

bias limits the theory's capacity to critically assess the negative ethical, social, or environmental impact of innovations such as GenAI (Kenthapadi et al., 2023). This limitation is significant in the higher education context, where the framing of innovation as inherently desirable can also silence critical discussions about whether technologies align with educational values or learning outcomes (Selwyn, 2013). Therefore, we used more neutral labels that acknowledge the informed skepticism rooted in legitimate concerns about AI. Nonetheless, the differences across three classes and their evolution over time remain consistent with the staged adoption process described by Rogers, and the implications for higher education are discussed in the following sections.

How Experiences Shape AI Perceptions

According to Rogers et al. (2014), the innovation-decision process begins with individuals becoming aware of the innovation and then forming a favorable or unfavorable attitude toward the innovation. Therefore, while initial positive attitudes can lead to adoption and use, the outcomes of that experience, in turn, can further influence attitudes, suggesting a reciprocal relationship. Our data shows that the composition of different classes varies between Year 1 and Year 2, yet the characteristics of each class remain consistent. Specifically, Advocates are more experienced with GenAI and, at the same time, more positive toward its usage in education scenarios. This aligns well with the descriptions of Innovators who are more open to exploring new technologies and affirming their usefulness within the community (Ghimire & Edwards, 2024). Moreover, the clear distinction in experience levels across three classes indicates that the increased experience, training, and education on GenAI are more likely to lead to a positive instead of negative perception of AI, which echoes the findings from prior research on the relationship between experience and perception of AI (Horowitz & Kahn, 2021; Kim & Lee, 2024).

Noticeably, in Year 1's results, regardless of the experience level, all classes agreed that students' use of GenAI in coursework would be inevitable. However, Skeptics are inclined toward restrictive policies that stop students' use of GenAI and view it as a violation of academic integrity. This inclination may represent the apprehensions and skepticism often associated with the initial adoption phases of disruptive technologies, which can stem from a combination of societal and cultural factors, such as the sense of losing control (Bostrom, 2014), lack of understanding (Kastania, 2024), and ethical concerns (Zohny et al., 2023). In educational settings, the introduction of new technologies has historically sparked concerns regarding academic integrity and the potential for diminished student learning (Aagaard, 2015; Mobilio et al., 2024; Roberts, 1980). Nonetheless, it is well established in the literature that fostering a positive and

supportive educational environment is preferable to punitive actions for addressing academic integrity issues (De Maio & Dixon, 2022). Therefore, recent publications have been advocating that the ethical integration of GenAI will not undermine but instead improve education (Tan & Maravilla, 2024), which is supported by preliminary evidence for pilot implementation of GenAI-supported assessment in higher education (Furze et al., 2024). Findings from the current study highlight the importance of increasing the level of experience and training with GenAI among faculty. This can mitigate the negative presupposition towards disruptive technology and facilitate a more supportive and educational approach to GenAI.

How AI Perceptions Evolve over Time

It should be noted that there was a substantial shift in sample composition between Year 1 (334 students, 256 faculty) and Year 2 (65 students, 80 faculty, 219 staff). This shift complicates direct comparisons across time because the reduced number of student and faculty participants may amplify the perspectives of a smaller subset of respondents, while the absence of staff members in Year 1 prevents comparison analysis for that group altogether. Nevertheless, the fact that similar categorizations and characteristics emerged, with three classes among students and faculty in Year 1, and two classes among faculty and staff in Year 2, suggests the possibility of evolving perspectives across the university community despite differences in sample composition. Specifically, for students in Year 2, there are only Explorers who are experienced with GenAI but also aware of its risks and concerns. The disappearance of Advocates and Skeptics, either due to their unwillingness to participate in the study or reduced proportion in the population, indicates that more students may have gained exposure to GenAI, but in a protected manner. Studies show that many universities have published guidelines for faculty and students, providing definitions and examples for trustworthy AI tools and appropriate AI usage (McDonald et al., 2024; Moorhouse et al., 2023). Meanwhile, the concern over being accused of plagiarism is an important factor in lowering students' intention to use AI tools (Ortiz-Bonnin & Blahopoulou, 2025). Despite concerns, the Explorers are still willing to receive more training in AI. Therefore, there is still a need for consistent university-wide training programs that help improve AI literacy among all students.

For faculty, we observed 43% of Advocates in Year 2. While the difference between Year 1 and Year 2 is possibly attributed to changes in sample compositions, the absolute value still indicates a significant number of faculty who are more familiar with AI. Moreover, despite discomfort with AI usage in various scenarios, Skeptics' agreement on integrating GenAI into instruction in the future has reached slightly

over neutral ($C1 = 3.2$) in Year 2. The two classes also form a consensus that students should be informed about AI usage in instructional support and that the university needs to teach them about responsible AI usage. These trends align with recent studies reporting the overall cautious yet optimistic stance of universities and the emphasis on faculty's responsibilities of integrating GenAI into curriculum and assessment (Cacho, 2024; Jin et al., 2024). However, despite the raised awareness that GenAI will be an indispensable part of higher education, the low experience level and strong interest in training among the Skeptics suggest a lack of effective AI literacy improvement programs during the period. In addition, viewing AI usage as a violation of academic integrity policies is still a key aspect that differentiates the two classes. Admittedly, GenAI's impact on academic integrity is a complicated topic that varies based on context (Yusuf et al., 2024). Existing recommendations include a more comprehensive policy and more adaptive assessments to accommodate the use of GenAI (Rasul et al., 2024), which would require changes in the faculty's pedagogical practices. Faculty who are less experienced with GenAI may perceive more challenges and workloads associated with these changes.

How Students, Faculty, and Staff Perceptions Align/Misalign

Involving multi-stakeholders in the processes of designing education technology solutions enhances inclusivity, transparency, and empowerment, leading to improved educational quality (Iqbal et al., 2024). Therefore, it's important to take a holistic approach to understand the potential impact of GenAI on students, faculty, and staff. Each of the groups has distinct experiences and responsibilities on campus, and our survey reveals nuanced differences among their perceptions of GenAI. In Year 1, there is a larger group of Skeptics among students, taking up 50% of respondents compared to 21% among faculty. This contradicts prior studies that observed students generally view GenAI more positively than faculty or instructors (Chan & Lee, 2023; Zastudil et al., 2023). Students could feel more uncertain about GenAI due to more strict university or classroom policies, which seemed true for Year 1, as faculty agree that the use of AI to complete coursework is cheating ($P3 = 3.66$) and a violation of academic integrity policies ($P4 = 3.66$). Students in Year 2 also showed intense concern over academic integrity issues and a lower level of comfort with AI usage in classrooms. Also noticeable is that across the years, students have low levels of agreement on instructors' successful use of GenAI. A lack of successful GenAI usage in instruction may contribute to a more negative view of AI among students.

Out of all the concerns associated with GenAI, the future job security concern is more evident for students than other

groups, echoing literature underscoring the fears about automation and AI-driven displacement among younger generations preparing to enter the workforce (Choi & Kang, 2019). This anxiety about future job security can lead to disengagement among students, as they may feel that their educational efforts are being undermined by the rapid adoption of AI in professional settings. As higher education is already experiencing issues with enrollment and student retention (Eather et al., 2022), it is critical to address the AI challenges appropriately. Faculty may need to teach students how to collaborate with AI effectively in writing or design tasks and shift their perspective towards seeing AI as a valuable skill in the workplace. Institutions should recognize students' fears and offer career guidance that highlights emerging job opportunities created by AI. Creating partnerships with industries that actively integrate AI into their workflows can help students gain firsthand experience with AI tools, reducing uncertainty and building confidence in their ability to adapt to AI-enhanced workplaces (Rane, 2023).

Non-academic staff also play important roles in supporting teaching, learning, and research activities at universities, and their perceptions of AI are critical for ensuring ethical, inclusive, and collaborative adoption of AI, which ultimately improves student outcomes. While we only surveyed staff perceptions in Year 2, the results show similar patterns to those of other groups. Specifically, we observed the existence of Advocates and Skeptics, which made up 66% and 34% of the staff responses, respectively. The larger proportion of Advocates suggested that staff may be more open to gaining experience with AI. One potential explanation is that they are less involved with the academic integrity controversies, as evidenced by their neutral attitudes toward whether AI usage violates academic integrity and whether they adhere to university policies on AI usage. Nonetheless, they are still aware of AI's risks in causing academic integrity issues, advancing bias, and compromising privacy. Staff members, especially IT personnel, often provide essential support to faculty in adopting and using digital technologies, and they can encounter challenges in unclear communications, limited resources, and resistance to change (Polly et al., 2021). Therefore, institutions need to provide timely resources and professional development for staff so that they can better support faculty and students in addressing GenAI issues effectively.

Limitations and Future Work

This study employed a primarily quantitative approach to explore the university community's attitudes toward GenAI and to identify latent attitudinal profiles through survey data. While the multifaceted questionnaire was designed to capture a wide range of perspectives, it was exploratory

in nature. The items were empirically derived rather than grounded in established theoretical frameworks, which limits their capacity to reliably measure underlying psychological constructs without further validation. Future research should validate and revise the items based on established theories such as TAM or UTAUT, to strengthen construct validity and facilitate comparisons across studies. Additionally, the study was conducted within a single institutional setting, shaped by the university's unique cultural, technological, and policy environment surrounding GenAI, which limits the generalizability of the findings. Future work could expand to multiple institutions with diverse profiles to test the transferability of findings and account for broader contextual influences.

Moreover, the dramatic shift in sample composition between Year 1 and Year 2 complicates claims about evolving perceptions over time, as differences in respondent characteristics may partly account for observed changes. Several factors may have contributed to the drop-off among students and faculty. Institutional priorities likely shifted between years, limiting opportunities for coordinated outreach and reducing the study's visibility among potential participants. In addition, at a large research institution, the high frequency of campuswide surveys can quickly lead to survey fatigue, lowering participants' willingness to engage. Notably, the number of studies focusing on GenAI grew substantially from 2023 to 2024, potentially diluting interest in any single survey on the topic. Future research should consider more robust sampling strategies to support evaluations of longitudinal change.

Qualitative methods such as interviews or focus groups could also enrich our understanding of the motivations and concerns underlying GenAI attitudes, offering deeper insight into themes surfaced through quantitative analysis. In addition, employing more advanced analytical approaches, such as structural equation modeling, could strengthen the evidence for patterns of technology adoption and the mechanisms driving attitudinal change. Overall, while this study offers a foundational view of GenAI perceptions in a university context, future research is needed to validate its instruments, broaden its scope, and deepen its insights.

Conclusion

This study expands our understanding of how GenAI is shaping higher education by examining the evolving perceptions of students, faculty, and staff. Through two years of survey data and latent class analysis, we identified distinct attitudinal profiles and shifts in GenAI perceptions across campus roles. While perceptions vary by experience and role, a clear pattern emerged: greater familiarity with GenAI tends to foster more positive attitudes and openness toward its use in education. The findings highlight the reciprocal

relationship between experience and perception, suggesting that targeted training and exposure can play a pivotal role in promoting thoughtful GenAI adoption. Institutions should prioritize comprehensive AI literacy initiatives that not only equip faculty with tools for ethical and innovative teaching but also support students in navigating AI's impact on career pathways and academic expectations. Ultimately, the successful integration of GenAI into higher education depends on inclusive, role-sensitive approaches that address the concerns and aspirations of the entire academic community.

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Declarations

Competing interests The authors declare no potential conflicts of interest for this research and received no outside funding for the study.

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